



Genoma

*Framing the gene editing problem.
Accelerating cures.*

A nonprofit dedicated to curing genetic disease.

The Scale of the Problem

1 in 10

people affected by
a genetic disease

400M

suffer globally from
a rare disease

95%

of rare diseases lack
an FDA-approved treatment

10,000+

known types of
rare diseases





46 approved gene therapies
= 0.23% of all human genes

We are only scratching the surface.



- Approved gene therapies (46)
- Known genetic disorders — no therapy (~9,954)
- Genes not yet linked to a known disorder (~10,000)

46

approved
gene therapies today

10,000

known
genetic disorders

20,000

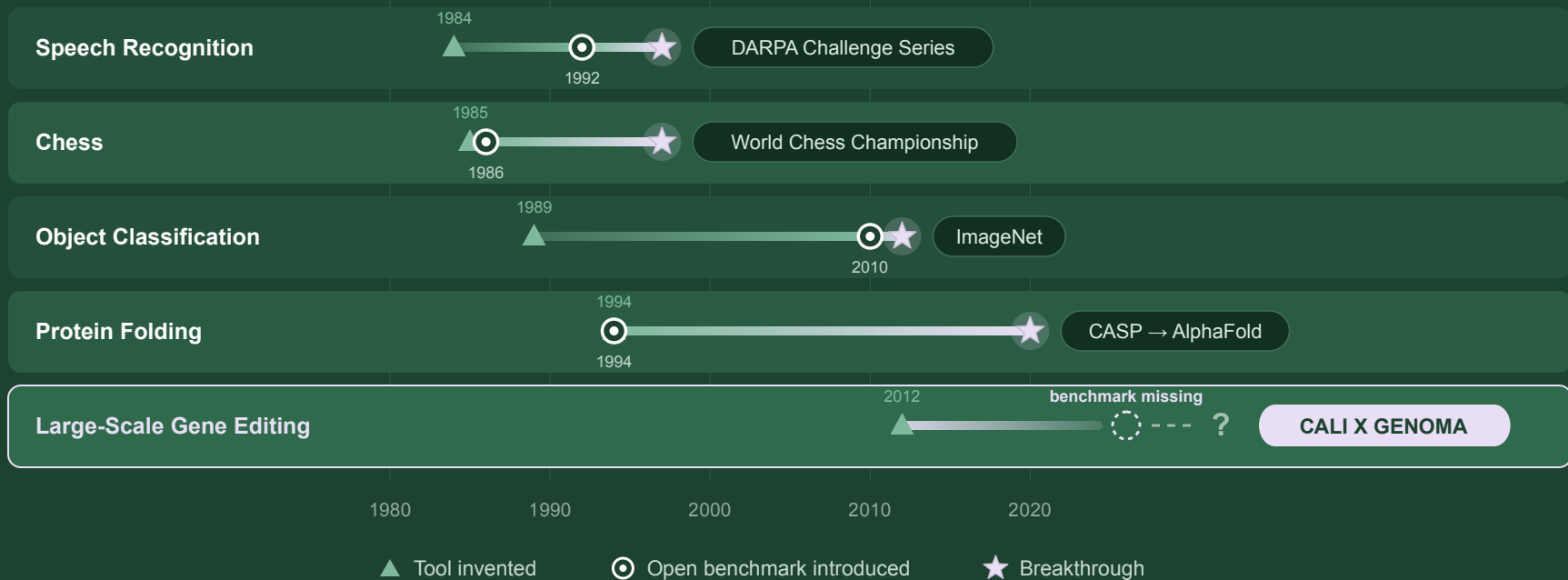
total protein-coding
genes in the genome

0.23%

of all human genes
have an approved therapy

History Shows How This Gets Solved

In every major field, the breakthrough came after someone framed the problem with an open benchmark.



The star only appears after the circle. Gene editing has the triangle. Genoma places the circle.





Genoma Institute

*A nonprofit dedicated to
curing genetic disease.*

Our mission is to build an **open community**
that **emphasizes scientific rigor** and
reproducibility to enable our vision which is to
cure genetic disease.

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01 CALI Competition

Critical Assessment of Large Cargo Insertion

A biennial, blind, unbiased competition, think **the Olympics of large-scale gene editing**. Benchmarks that let academia and industry compete, compare, and learn.

02 Research Lab

Mitochondrial Gene Therapy

A focused nonprofit lab developing gene therapies for mitochondrial DNA diseases, tackling age-related and rare disorders no one else is chasing.

03 State of the Genome

Public Science Communication

A newsletter, podcast, and forum creating a trusted, structured space for public discourse on gene editing, AI in biology, and what it means for humanity.

Dr. Lisa Eshun-Wilson is uniquely positioned to bring this to life



Structural biologist with **10+ years of cryo-EM expertise**, spanning microtubules, viral glycoproteins, and mitochondrial AAA+ proteases. **Two co-first-author Science papers**, a first-author *PNAS* paper, and 750+ citations. Designed programmable DNA recombinases at Profluent Bio, Inc., an AI-first protein design company. **NSF (GRFP, PRFB) and Ford Foundation Fellow**; invited speaker in Germany, Hong Kong, Singapore, and Saudi Arabia.

2010

Grinnell College | B.A. Biological Chemistry · Where the curiosity for enzymology began

2014

NIH/NHLBI | Membrane Biology, Cellular Trafficking, & Endocytosis · Dr. Julie Donaldson

2015

UC Berkeley | Ph.D. Biochemistry, Biophysics & Structural Biology · Microtubule regulation with cryo-EM

2018

inclusiveBio | Founded at Berkeley · Raised \$100K+ · Built 300+ person community

2020

Scripps Research | Mitochondrial health and regulation with Dr. Gabe Lander using cryo-EM

2024

Profluent Bio | AI-first protein design of programmable DNA recombinases · The exact tools CALI will benchmark

Now

Genoma Institute | Building the open benchmark for programmable gene insertion



CALI | Critical Assessment of Large Cargo Insertion

The Olympics of Gene Editing



Inspired by CASP, the competition that gave us AlphaFold and a Nobel Prize. CALI will do the same for gene therapy.



State of the Genome | Public Science Communication

A trusted, structured space for the most important conversations in biology.

Newsletter

Accessible, science-first breakdowns of gene editing advances. Written for curious, educated readers.

Podcast

Deep conversations with scientists, ethicists, patients, and policymakers. Long-form, nuanced, unsponsored.

Forum

A structured public space for discourse on scientific progress, setbacks and experimental challenges in the field.

Why this matters now

There is enormous public anxiety around AI, designer babies, and genetic racism — and almost no trusted scientific voice in the space. State of the Genome fills that gap. It protects the science from misinformation and brings the public into the conversation before policy is made without them.





Partner With Us

Your support funds the inaugural CALI competition, the first open benchmark in gene therapy since CASP transformed protein folding.

Founding Partner

\$500K+

Program Partner

\$100–500K

Community Partner

Up to \$100K
